National Telecommunication Institute

وزارة الاتصـــــالات
وتكنولوجيا المعلومات

MSSP

MANAGED SECURITY SERVICE PROVIDER)

MCSA

The Marshmallo project represents an integrated infrastructure for a company specialized in providing MSSP (Managed Security Service Provider) services.

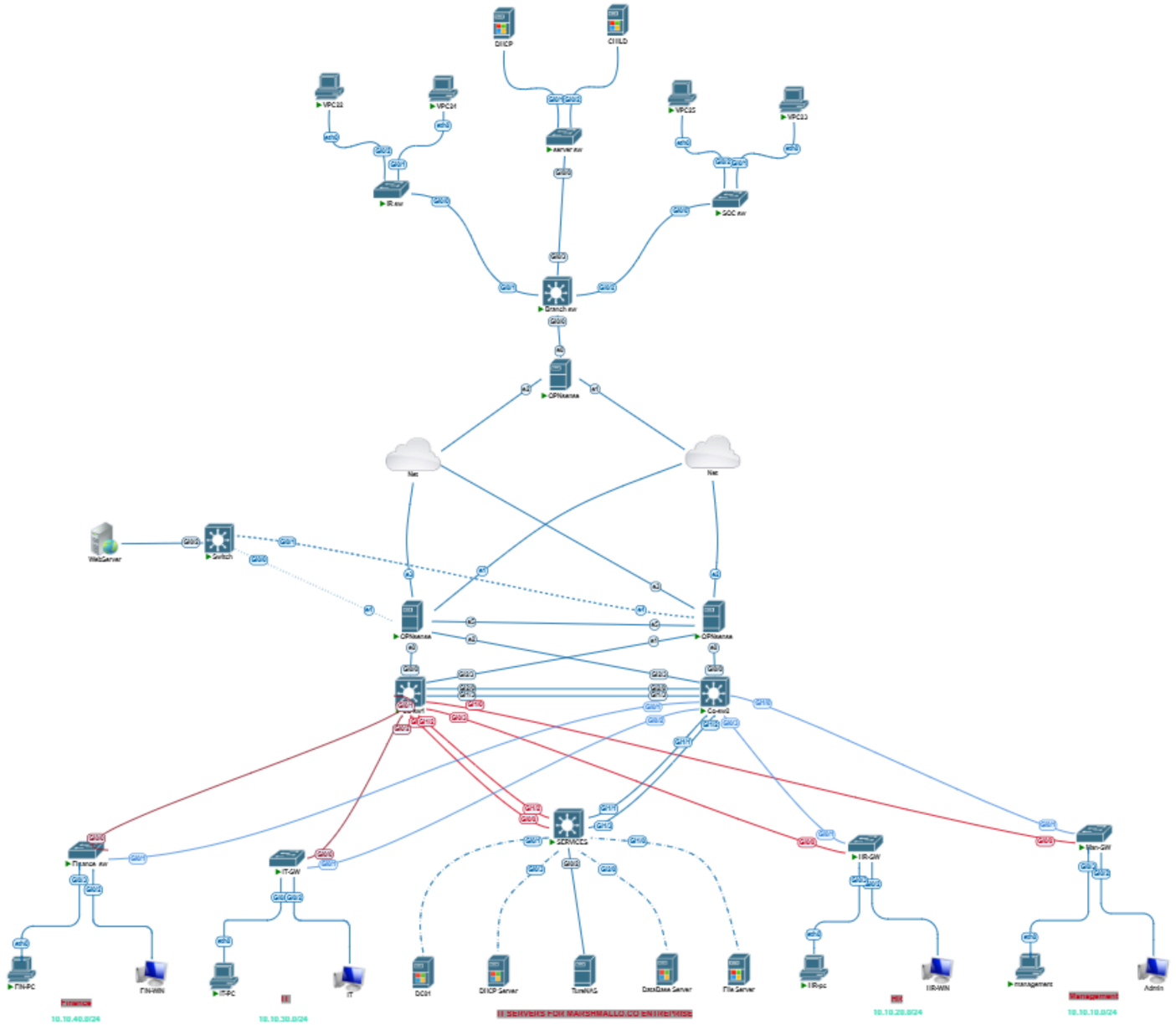
Project Overview

Company Overview – Marshmallo

Marshmallo is a Managed Security Service Provider (MSSP).
Provides Cybersecurity & Network Security services for enterprises.
Secures enterprise infrastructure and monitors network traffic centrally.
Uses security solutions such as:

Next-Generation Firewalls (NGFW)
SIEM
IDS/IPS
Secure Remote Access
Provides centralized monitoring,
threat detection,
and security management.

TOPOLOGY



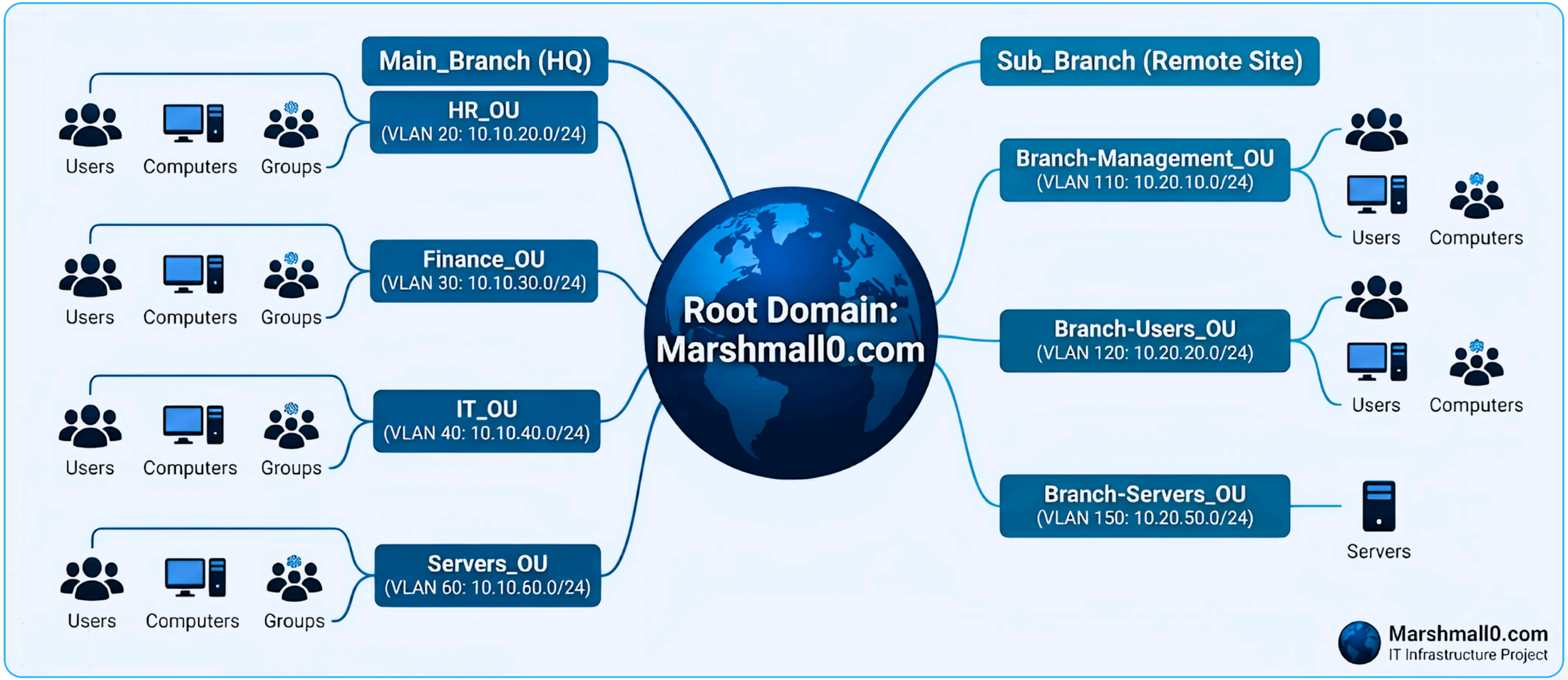
HQ

- DC
- DHCP
- TureNAS
- DataBase Server
- FileServer
- WebServer

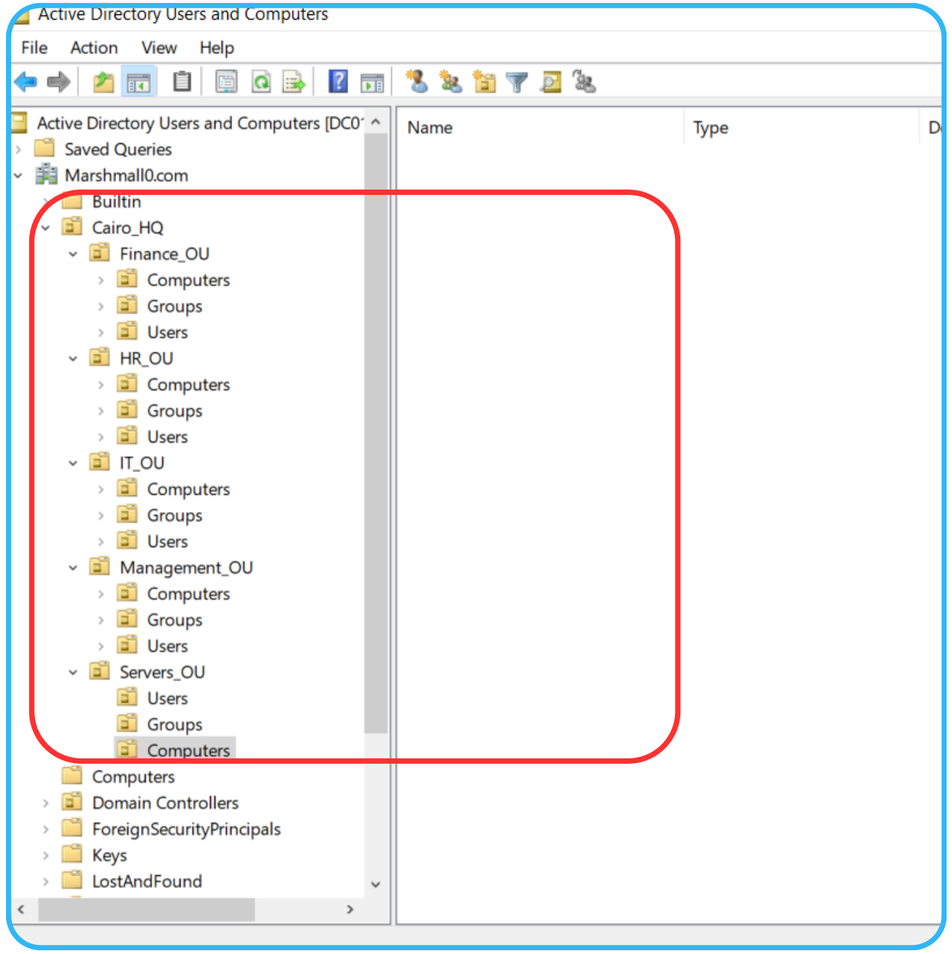
Branch

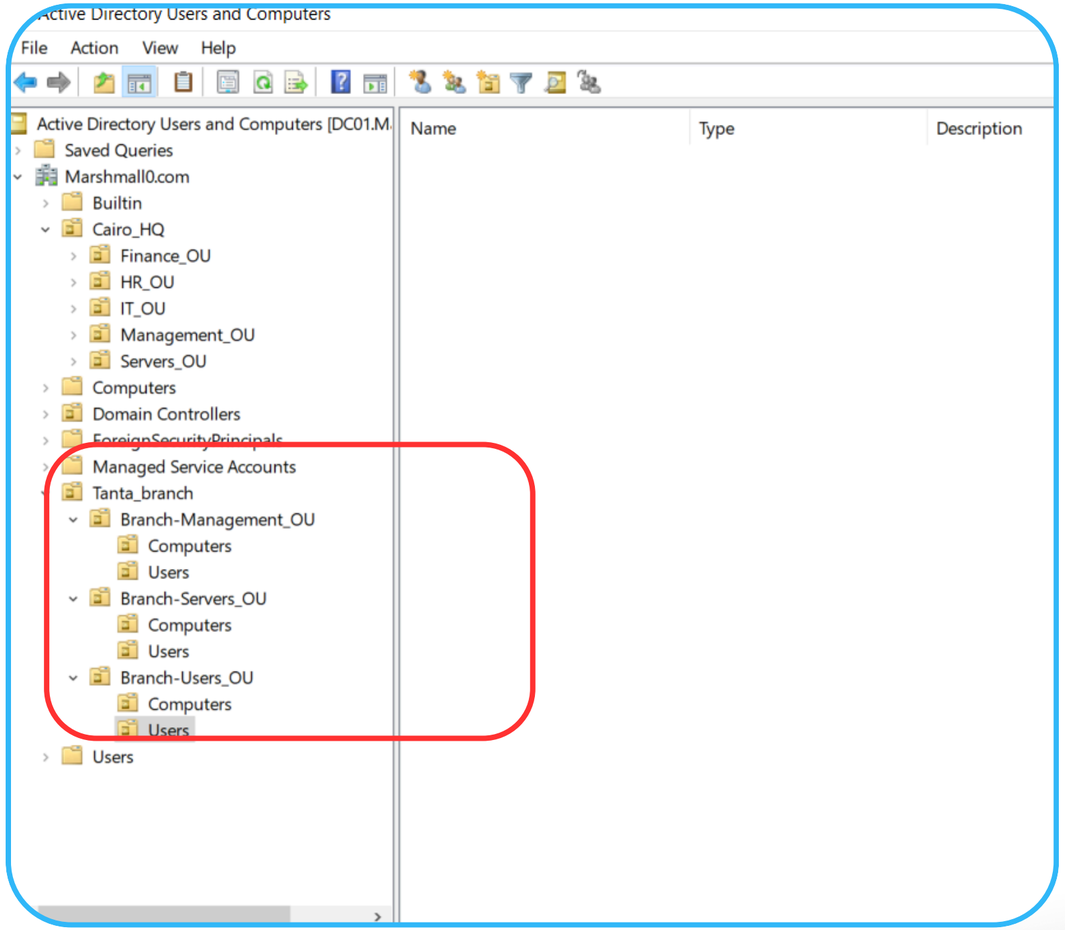
- childServer
- Dhcp_Branch

Active Directory OU Architecture

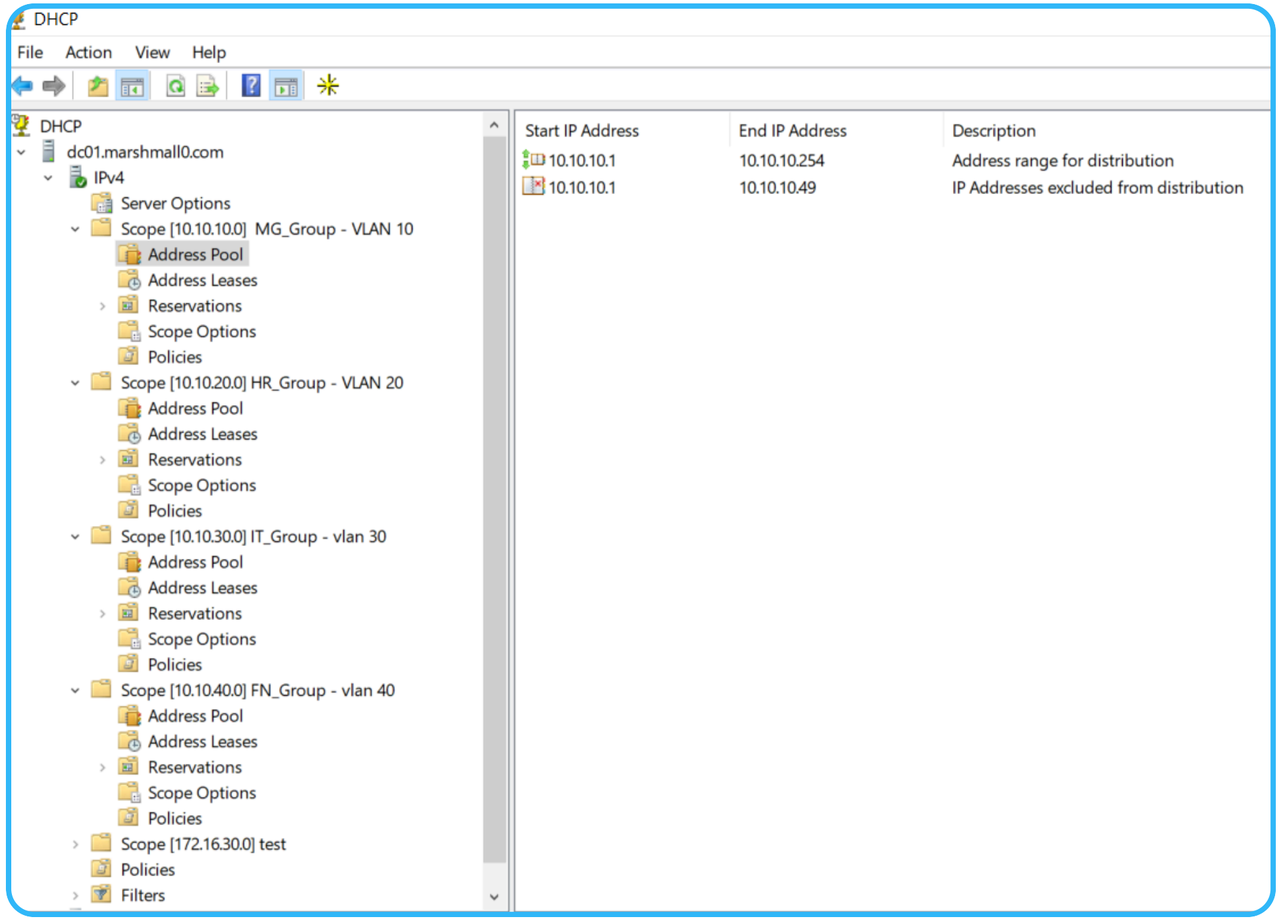


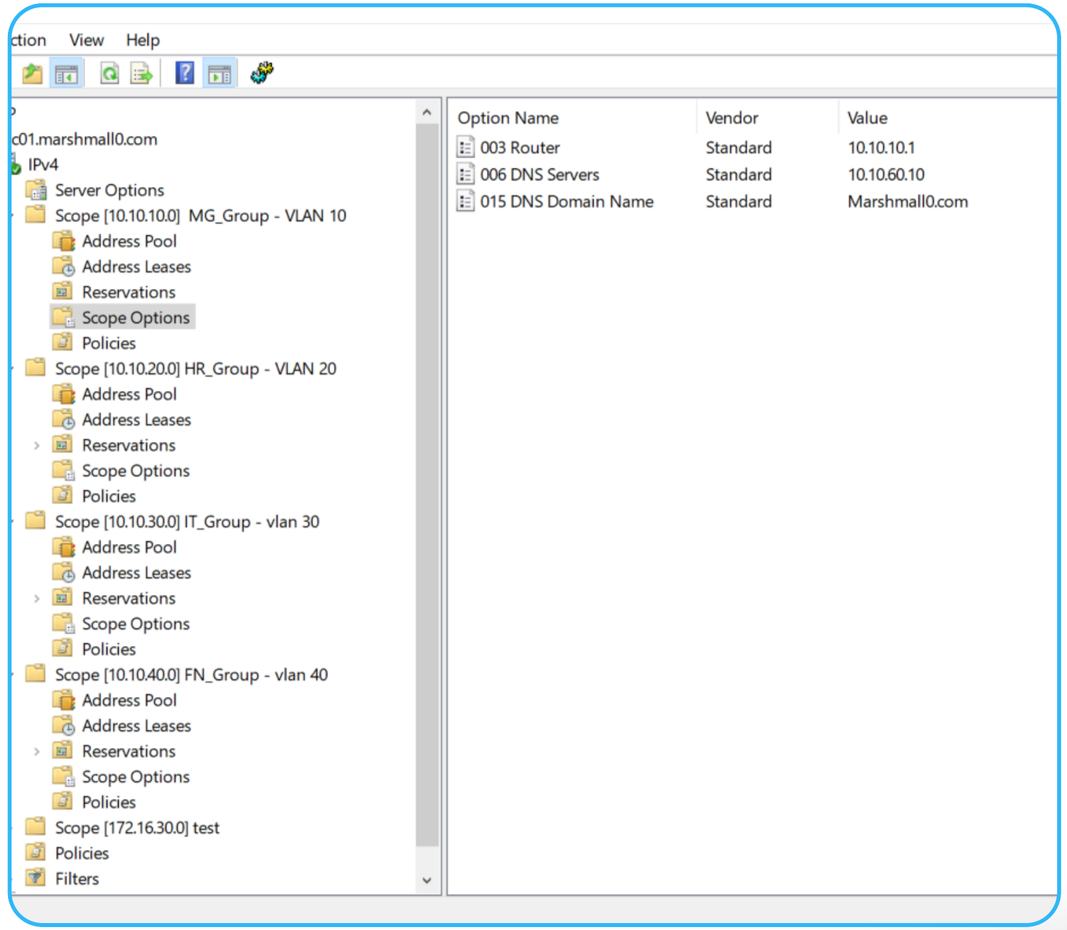
Active Directory OU Architecture



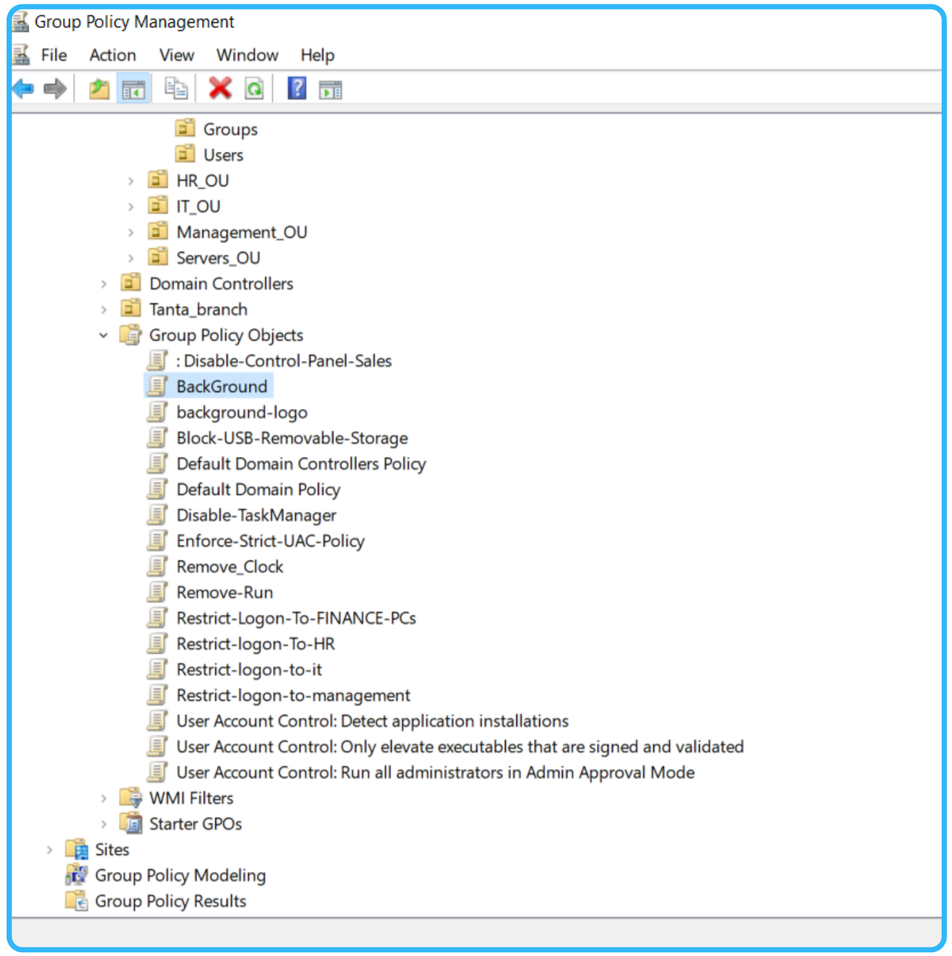


Active Directory OU Architecture





GPO



GPOs – Main Functions

Disable-Control-Panel-Sales → Blocks Control Panel for Sales.

BackGround → Applies standard desktop background.

background-logo → Applies company logo.

Block-USB-Removable-Storage → Blocks USB storage devices.

Default Domain Controllers Policy → Basic DC security settings.

Default Domain Policy → Basic domain-wide policies.

Disable-TaskManager → Blocks Task Manager.

Enforce-Strict-UAC-Policy → Strengthens UAC security.

Remove-Lock → Removes the Lock option.

Remove-Run → Blocks the Run command.

Restrict-Logon-To-FINANCE-PCs → Limits Finance logon.

Restrict-logon-To-HR → Limits HR logon.

Restrict-logon-to-it → Limits IT logon.

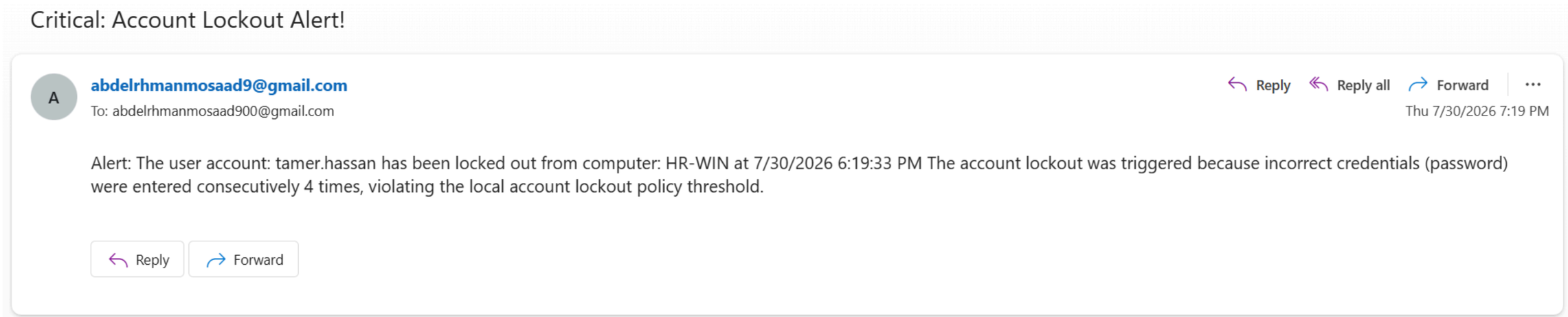
Restrict-logon-to-management → Limits Management logon.

UAC – Detect Application Installations → Detects application installations.

UAC – Signed & Validated Executables → Allows only trusted signed apps.

UAC – Admin Approval Mode → Requires admin approval for elevated actions.

Automated Email



Automated Account Lockout Email Alert

1-Detect Lockout Event → Monitor Security Event ID 4740.

2-Create PowerShell Script → Detects the locked account and sends an email alert.

3-Configure SMTP → Set sender, administrator email, SMTP server, and authentication.

4-Save Script → Save the PowerShell file as LockoutAlert.ps1.

5-Test the Script → Run the .ps1 file as Administrator.

Configure Task Scheduler:

Create a new task: Lockout_Email_Alert.

Trigger: On an event → Security → Event ID 4740.

Action: Start powershell.exe.

Run the LockoutAlert.ps1 script.

6-Final Test → Trigger an account lockout and verify that the administrator receives the email alert.

Result: 🔔 Automatic email notification whenever a user account is locked out.

TrueNAS

TrueNAS is a centralized storage server that pools multiple hard drives and delivers high-performance block storage over the network to client servers.

Key benefits in short:

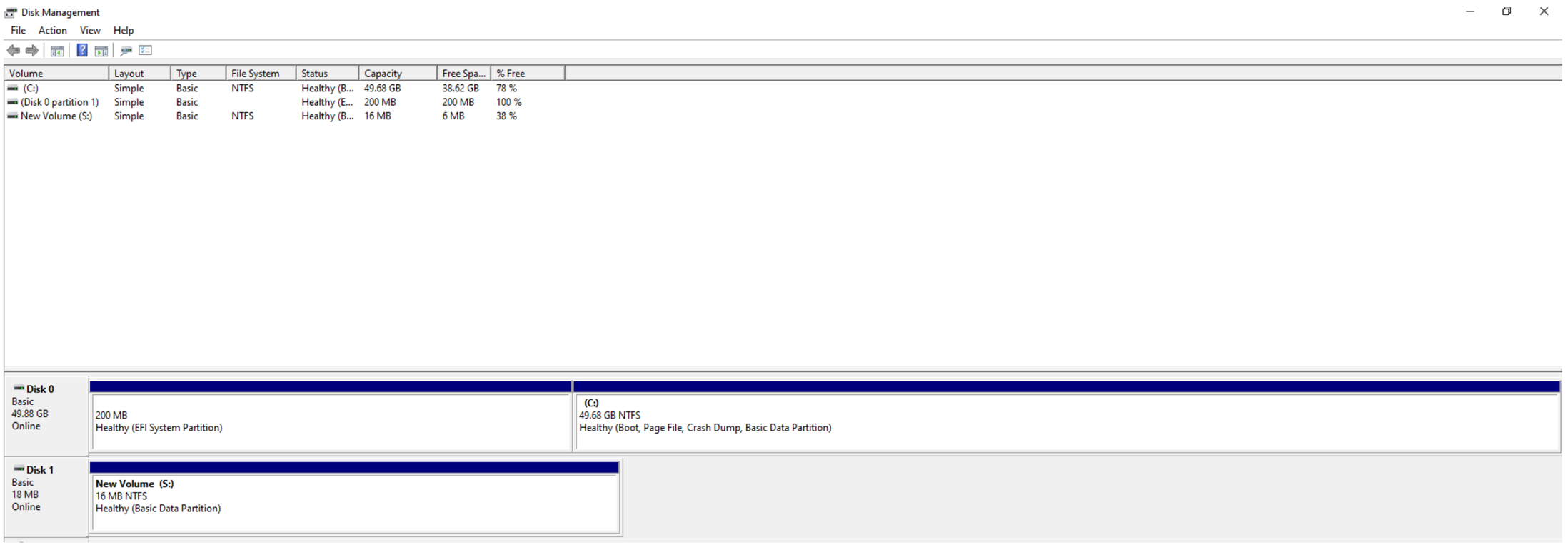
Centralized Storage: A single, massive place to store company databases and files.

Decoupling Storage from Compute: Other servers (like the database server) focus purely on processing work, leaving the heavy lifting of data storage to TrueNAS.

High Security and Data Protection: It protects data from corruption and provides snapshots and backups so you can easily recover your work if anything goes wrong.



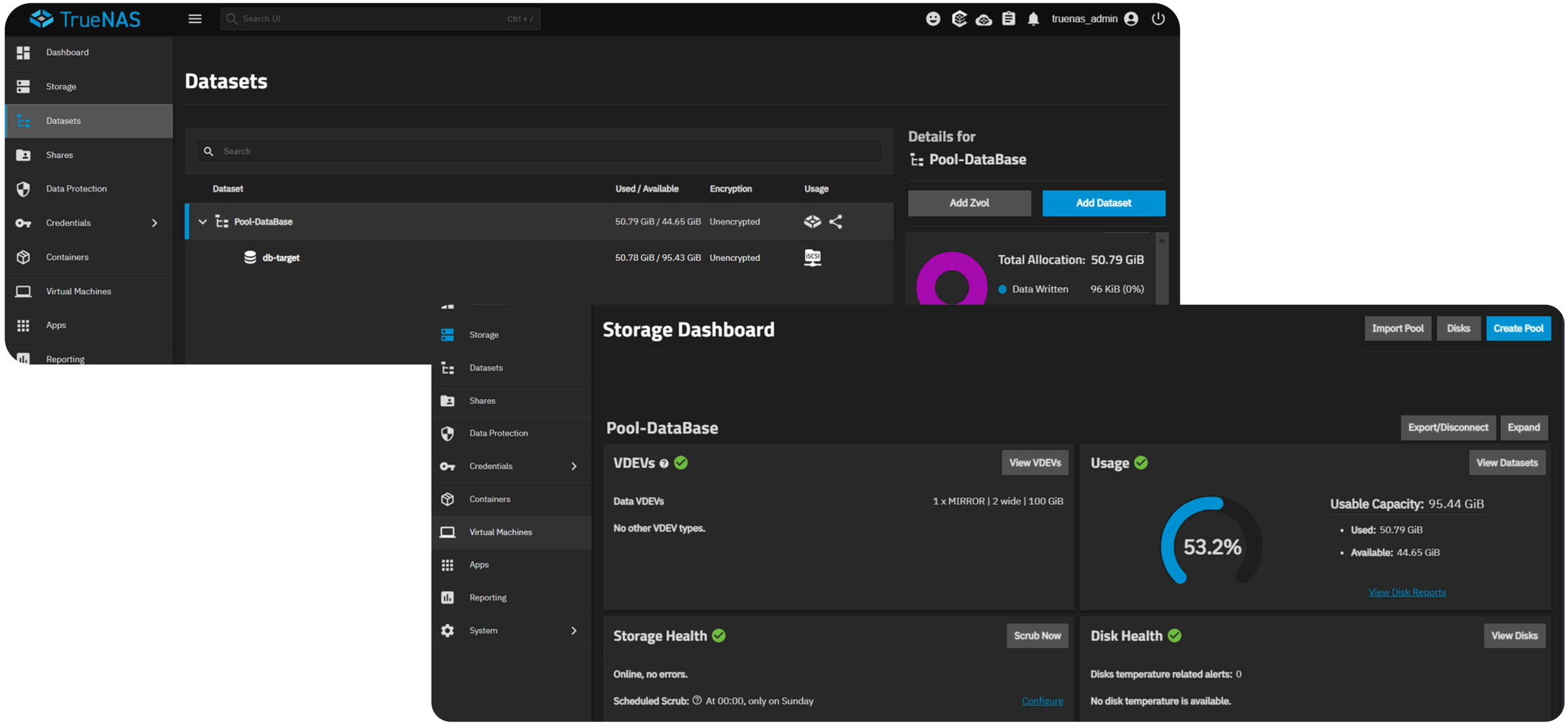
DataBase Server



iSCSI Initiator is the client-side component (like a Windows Server) that connects over the network to access and mount remote block storage provided by a TrueNAS Target, treating it as a local physical hard disk.

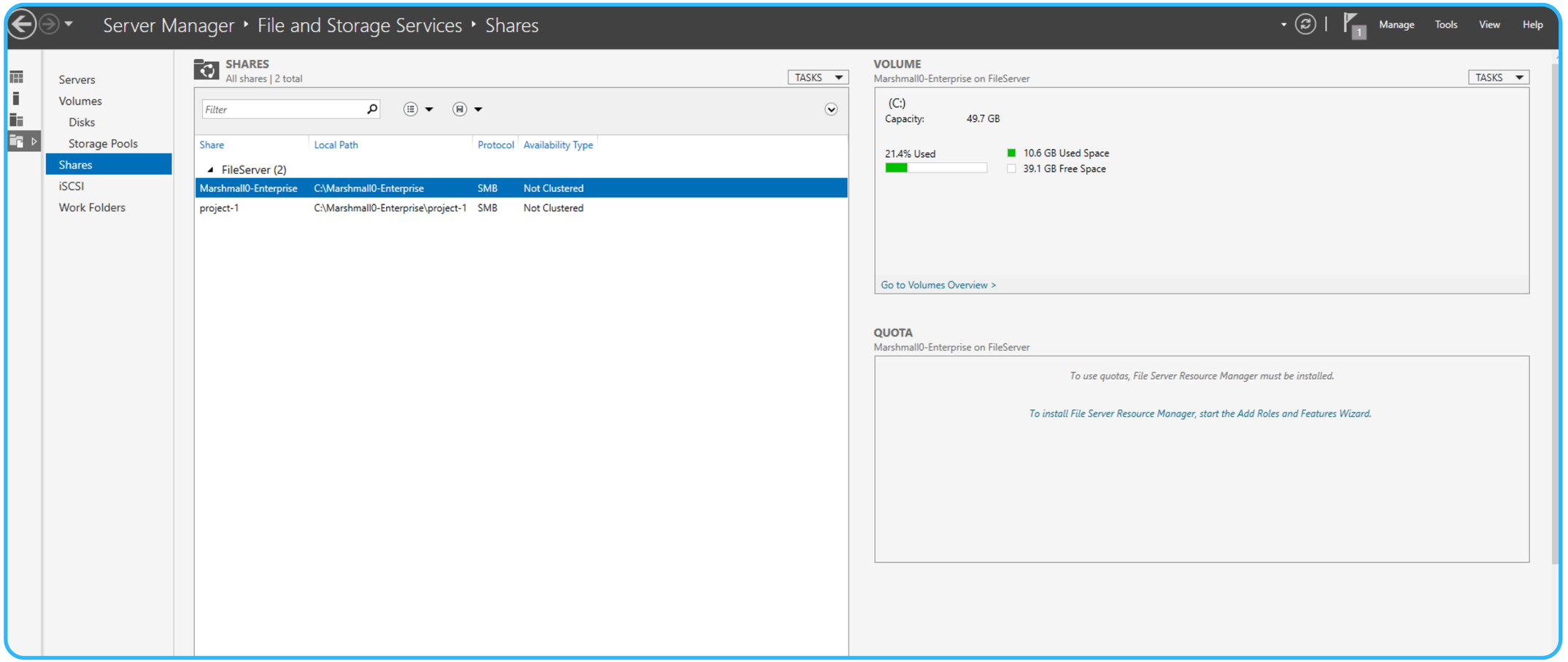
TrueNAS

TrueNAS Storage Dashboard & Pool Overview & CPU Resource



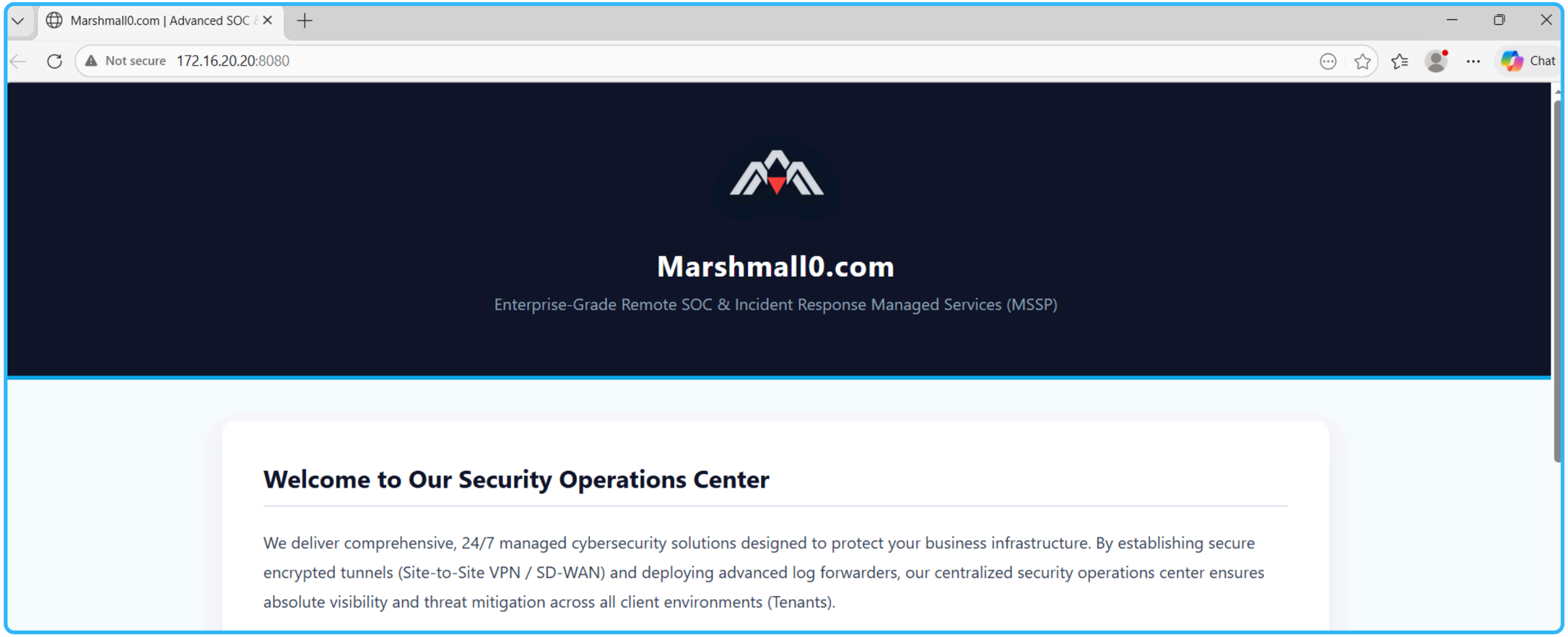
File Server

Core Concept: Centralizing company data storage on a dedicated server joined to the Active Directory domain to eliminate local data fragmentation and security vulnerabilities.



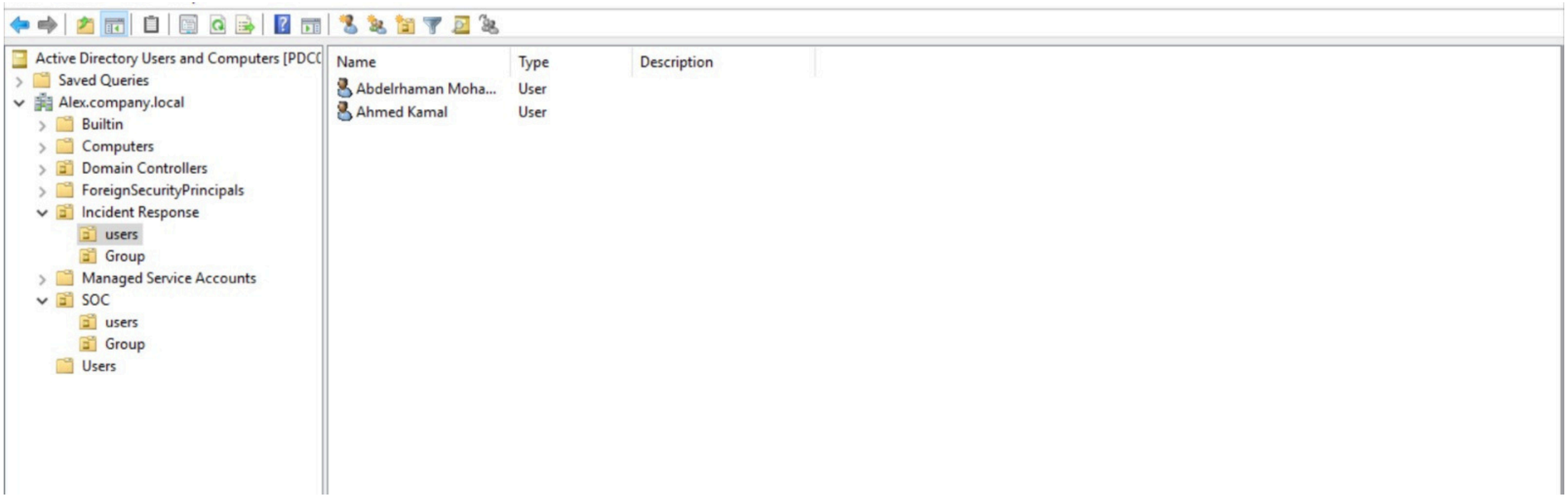
WebServer

Core Concept: Centralizing company data storage on a dedicated server joined to the Active Directory domain to eliminate local data fragmentation and security vulnerabilities.

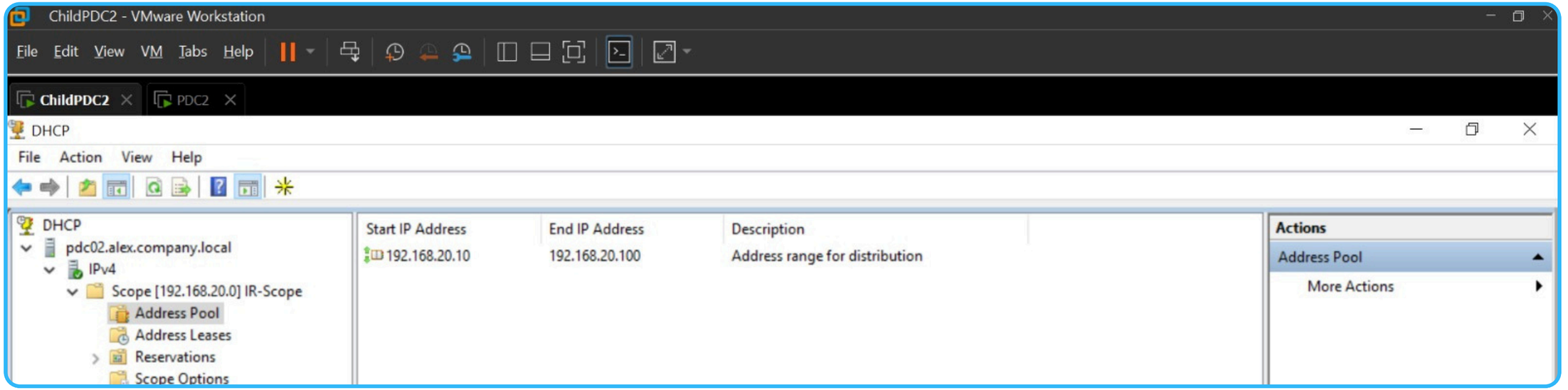


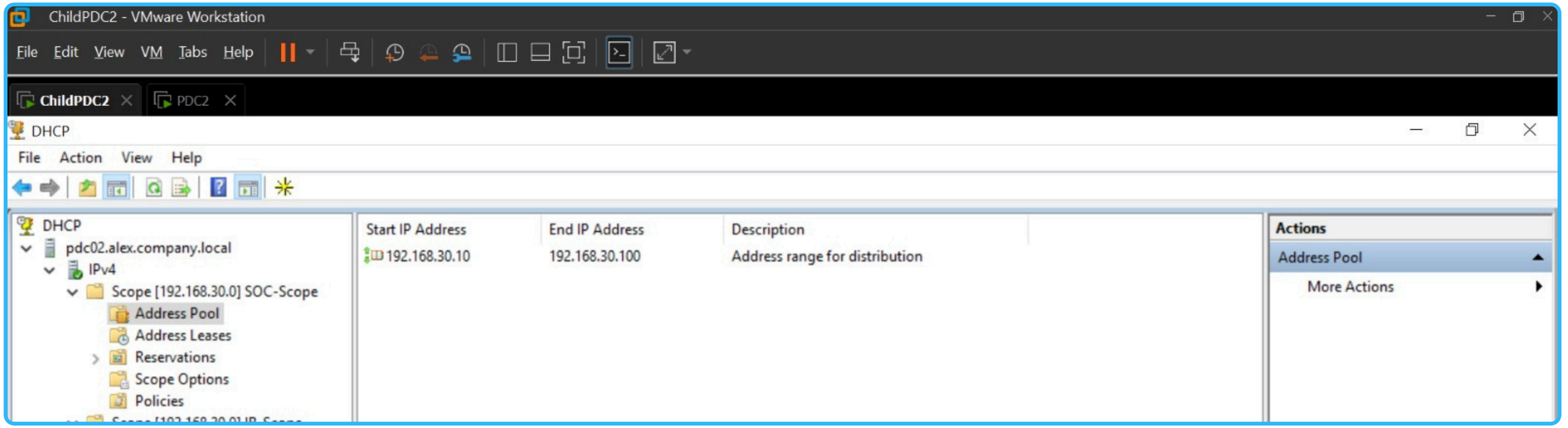
Child Server

Core Concept: Centralizing company data storage on a dedicated server joined to the Active Directory domain to eliminate local data fragmentation and security vulnerabilities.

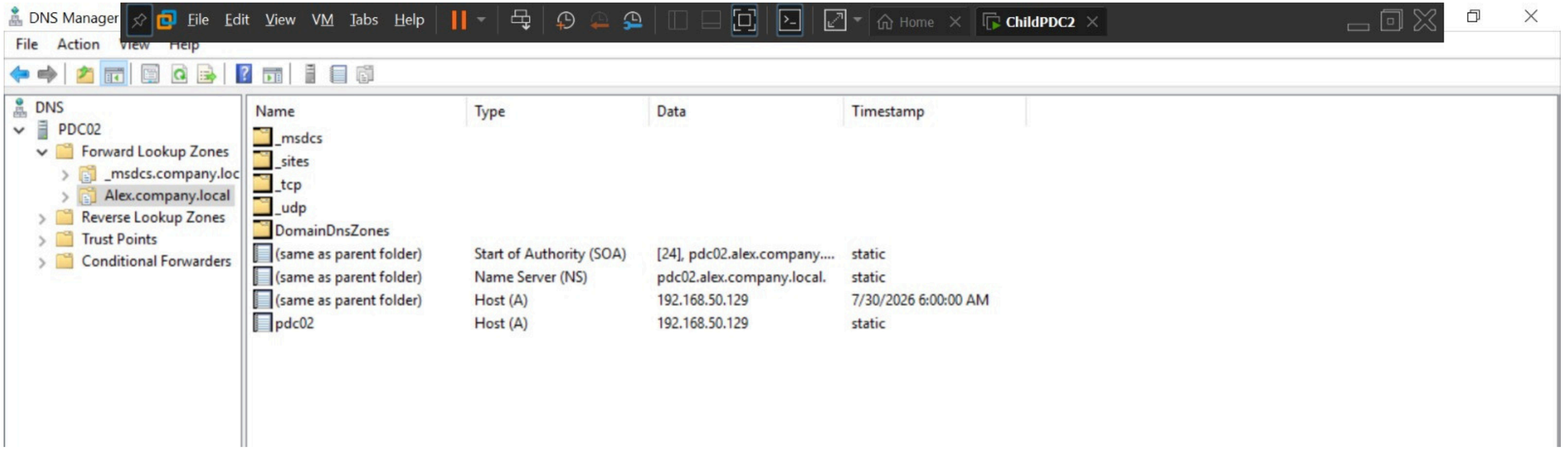


DHCP


ChildPDC2 - VMware Workstation
File Edit View VM Tabs Help
ChildPDC2
PDC2
DHCP
File Action View Help
DHCP
pdc02.alex.company.local
IPv4
Scope [192.168.20.0] IR-Scope
Address Pool
Address Leases
Reservations
Scope Options
Start IP Address
End IP Address
Description
192.168.20.10
192.168.20.100
Address range for distribution
Actions
Address Pool
More Actions


ChildPDC2 - VMware Workstation
File Edit View VM Tabs Help
ChildPDC2
PDC2
DHCP
File Action View Help
DHCP
pdc02.alex.company.local
IPv4
Scope [192.168.30.0] SOC-Scope
Address Pool
Address Leases
Reservations
Scope Options
Policies
Start IP Address
End IP Address
Description
192.168.30.10
192.168.30.100
Address range for distribution
Actions
Address Pool
More Actions

DNS


DNS Manager
File Edit View VM Tabs Help
Home
ChildPDC2
File Action View Help
DNS
PDC02
Forward Lookup Zones
_msdcs.company.loc
Alex.company.local
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name
Type
Data
Timestamp
_msdcs
_sites
_tcp
_udp
DomainDnsZones
(same as parent folder)
Start of Authority (SOA)
[24], pdc02.alex.company....
static
(same as parent folder)
Name Server (NS)
pdc02.alex.company.local.
static
(same as parent folder)
Host (A)
192.168.50.129
7/30/2026 6:00:00 AM
pdc02
Host (A)
192.168.50.129
static